Test-Time Compute and Semantic Safety Filtering for Industrial Fault Attribution

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***Abstract*—Industrial predictive-maintenance frameworks are typically validated under matched-severity conditions, masking a critical failure mode: classifiers trained on incipient faults collapse on the same fault type at advanced severity. We construct a load-matched cross-severity benchmark on the CWRU bearing dataset (training on 0.007-inch faults, testing on 0.021-inch faults) and show that SVM, Random Forest, and KNN classifiers all collapse on Inner Race recall (11 to 26 percent), indicating a feature-distribution problem rather than a classifier-choice problem. We then evaluate a Test-Time Compute (TTC) framework that performs fault attribution through self-consistent chain-of-thought reasoning over severity-invariant time-domain features. With five reasoning chains, the framework reaches 84.7 percent accuracy (F1 macro 0.845, an improvement of 17.3 points over Random Forest) and recovers Inner Race recall from 14 to 84 percent, trading 28 points of Outer Race recall for the previously uncorrectable bottleneck. For deployable safety, a sentence-transformer bi-encoder shield blocks 93.3 percent of a 30-case adversarial test set spanning direct, paraphrased, and obfuscated unsafe recommendations, versus 16.7 percent for word-matching, with a 0.4 percent false-positive rate on real outputs. We additionally report preliminary observations on consensus-pseudo-label adaptation of a 0.5-billion-parameter model and identify systematic small-model adaptation under richer training data as future work.**

***Index Terms*—Test-Time Compute, large language models, predictive maintenance, cross-severity generalization, bearing fault diagnosis, semantic safety filtering**

I. Introduction

Bearing failures are a leading cause of unplanned downtime in industrial rotating machinery. Recent work has demonstrated that large language models (LLMs) can be applied to textualized vibration features for fault classification [1], [2], raising the prospect of replacing bespoke classifiers with general-purpose reasoning systems. Three obstacles separate current LLM-based diagnostic frameworks from deployable industrial systems.

First, almost all reported results are obtained under matched-severity conditions, hiding the failure mode that matters most in practice: degradation as fault severity progresses past the training distribution. Second, no LLM-based industrial framework we are aware of validates the safety of its own natural-language outputs, and downstream pattern-matching filters break under simple paraphrasing. Third, existing systems treat fault diagnosis as classification rather than attribution: they emit labels without the physical reasoning that engineers would need to act on the output with confidence.

This paper addresses all three. We construct a load-matched cross-severity benchmark on the CWRU bearing dataset, evaluate a Test-Time Compute (TTC) framework that performs fault attribution through self-consistent chain-of-thought reasoning over physically grounded features, and pair it with a sentence-transformer semantic shield evaluated on adversarial unsafe recommendations.

II. Related Work

A. LLM-Based Bearing Fault Diagnosis

BearLLM [1] unifies multiple bearing-health tasks by adaptively sampling raw vibration signals, frequency-aligning them against fault-free reference recordings, and projecting the residual features into the embedding space of a LoRA-fine-tuned Qwen2-1.5B model. On the multi-dataset MBHM benchmark (which aggregates CWRU, MFPT, and seven other corpora), BearLLM reports 99.02 percent accuracy. The framework's reliance on a fault-free reference recording under identical operating conditions, however, makes it inoperable on devices whose conditions vary continuously, and all reported accuracies are obtained at matched severity.

Tao et al. [2] take an alternative textualization route, extracting 24 statistical features from raw signals and embedding them in natural-language prompts for a LoRA-fine-tuned ChatGLM2-6B. Evaluated across CWRU, MFPT, JNU, and PU, the method achieves an approximately 10-point cross-dataset improvement over non-LLM baselines. Like BearLLM, evaluation is offline and matched-severity; no mechanism is provided for post-deployment adaptation. A broader multi-modal framework by Alsaif et al. [3] reaches 96.3 percent on industrial-control fault records by combining retrieval-augmented generation with offline fine-tuning, but the authors explicitly acknowledge hallucinated outputs and conclude that human-in-the-loop validation remains indispensable.

A systematic re-evaluation of the CWRU dataset by Smith and Randall [4] assigns a P1 (partial-diagnosis) rating to record 222.mat—the 0.021-inch ball-fault file used in our cross-severity test track—because the envelope spectrum reveals genuine BPFI/BPFO cross-contamination at this severity level. This signal-level ambiguity is independent of classifier choice

and has not been acknowledged by any prior LLM-based diagnostic framework.

B. Test-Time Compute and Self-Consistency

Snell et al. [5] establish that allocating additional inference-time compute through sample-and-rerank or self-consistency procedures can match the accuracy of much larger fine-tuned models on mathematical reasoning, identifying a scaling regime in which test-time compute is preferable to additional pretraining. TTRL [6] extends this to weight-level adaptation, generating multiple candidate outputs on unlabeled data, treating the majority vote as a pseudo-label, and applying GRPO reinforcement learning. On Qwen2.5-Math-7B, TTRL improves AIME 2024 pass-at-1 from 12.9 percent to 40.2 percent. Critically, all such validations are confined to symbolic mathematical reasoning; the authors of TTRL explicitly identify continuous industrial time-series data as out-of-scope future work.

C. Safety in Autonomous and LLM-Based Systems

Verification-guided shielding [7] addresses safety in deep reinforcement learning by combining offline formal verification with selective runtime overrides; the resulting shield maintains a zero-collision rate while activating on only 1.3 percent of agent steps. The method assumes access to a formal model of environment dynamics and has not been applied to LLM-generated natural-language action spaces. Pattern-matching filters over curated unsafe vocabularies remain the dominant approach in deployed LLM systems, but—as we show in Section V—they are easily evaded by paraphrasing.

D. Research Gap

No prior framework jointly addresses cross-severity evaluation, deployable safety filtering for LLM-generated recommendations, and physically grounded chain-of-thought attribution on standard bearing benchmarks. This work targets all three.

III. Methodology

A. Framework Overview

The proposed framework operates as a four-stage inference pipeline.(Fig. 1): feature extraction (Stage 1), classical-classifier prior (Stage 2), Test-Time Compute reasoning with majority voting (Stage 3), and semantic safety shield (Stage 4). The shield post-processes natural-language outputs independently of the attribution result.

B. Feature Extraction

Each vibration segment is a 1024-sample window of drive-end accelerometer data, sampled at 12 kHz with 50 percent overlap. From each segment we extract 20 features: 8 time-domain statistics (RMS, peak, crest factor, kurtosis, skewness, variance, shape factor, impulse factor), 4 raw-spectrum descriptors (dominant frequency, dominant magnitude, spectral centroid, spectral spread), and 8 envelope-spectrum band amplitudes corresponding to the bearing characteristic frequencies BPFO, BPFI, BSF, FTF, and the physically meaningful sidebands BPFI$\pm f_r$ and BSF$\pm f_{\mathrm{FTF}}$.

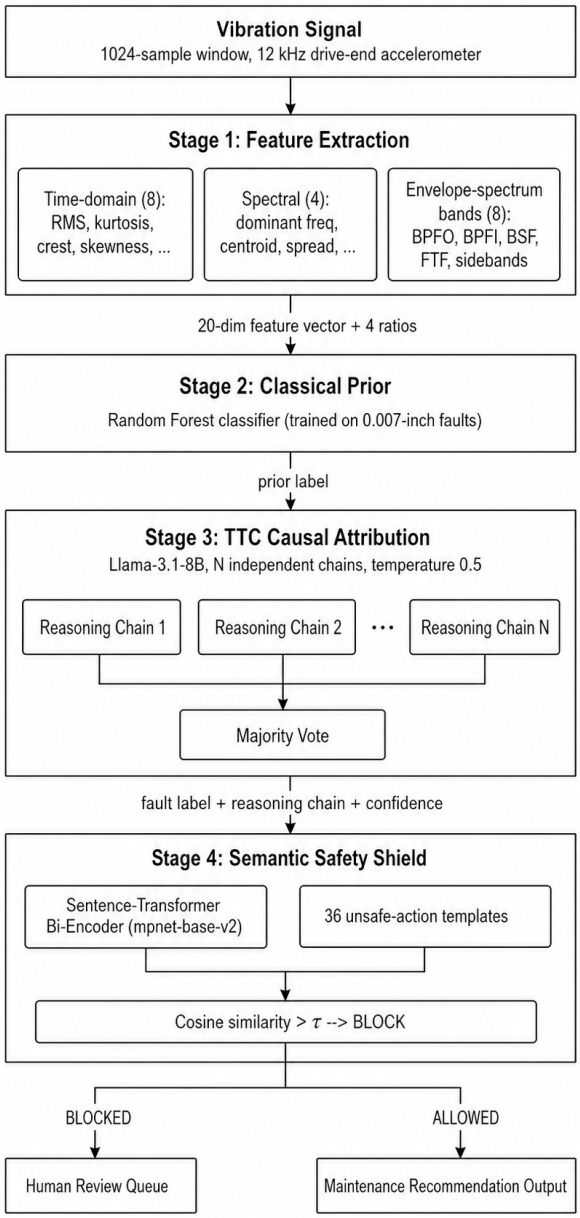

Fig. 1. Four-stage inference pipeline. Vibration features feed both a classical prior and the TTC reasoning layer. The semantic shield post-processes natural-language outputs independently of the attribution result.

The envelope-spectrum pipeline follows Method 1 of Smith and Randall [4]: the raw signal is bandpass-filtered, the analytic signal is computed via the Hilbert transform, the squared envelope is taken, and its Fourier transform yields the envelope spectrum. The narrow-band amplitude at each characteristic frequency is read as the peak magnitude within a ± 8 Hz window centred on the theoretical frequency. We additionally derive four spectral ratios (BPFI/BPFO, BPFI/BSF, BPFO/BSF, BPFI-sideband/BPFI) that are amplitude-invariant and therefore more robust under severity shift than absolute band energies.

We discuss a subtle implementation pitfall—using peak

amplitude rather than band-integrated energy—in Section V.

C. TTC Causal Attribution

Given a feature vector and a classical prior label, the TTC layer issues $N \in \{1, 3, 5\}$ independent reasoning chains from the same LLM, sampled at temperature 0.5. The system prompt specifies four ordered decision rules grounded in the empirical class statistics observed at advanced severity:

- Rule 1: If RMS < 0.25, classify as Ball (regardless of kurtosis).
- Rule 2: If kurtosis < 9, classify as Inner Race.
- Rule 3: If kurtosis > 10, classify as Outer Race.
- Rule 4: If kurtosis is in $[9, 10]$, use BPFI/BPFO > 1.0 as the Inner-versus-Outer tiebreaker.

Rules treat time-domain features as primary discriminators with spectral ratios as boundary tiebreakers. The model must agree with or override the prior, citing the specific feature value triggering a different rule. This symmetric design replaced an earlier asymmetric variant that anchored the model toward Inner Race irrespective of feature evidence (see Section VI.A).

Each reasoning chain emits a JSON object with three fields: a free-text reasoning chain naming the fired rule and triggering feature value, a fault label, and a confidence score in $[0, 1]$. The final attribution is the majority vote across N chains. We use Llama-3.1-8B served via Cerebras Cloud, with a 28 requests-per-minute token-bucket rate limiter.

D. Semantic Safety Shield

The shield is a sentence-transformer bi-encoder that embeds the natural-language recommendation produced by the framework and computes cosine similarity against a curated bank of 36 unsafe-action templates spanning four categories: continued operation, dismissal of fault, deferred maintenance, and stress escalation. Templates are paraphrased to include engineering-vocabulary variants ("the asset retains operational headroom"; "Bayesian updating against the prior of nominal bearing health does not yield a posterior sufficiently shifted"). A recommendation whose maximum cosine similarity to any template exceeds an operating threshold τ is blocked.

We use the all-mpnet-base-v2 sentence transformer [8], which produces 768-dimensional embeddings. The threshold τ governs an explicit trade-off between adversarial coverage and false-positive rate on legitimate maintenance language; we report the operating curve in Section V and select an operating point empirically.

IV. Experimental Results

A. Dataset and Experimental Setup

We use the CWRU dataset [4]: drive-end accelerometer signals from an SKF 6205-2RS bearing at 12 kHz, 1797 RPM, 0 HP load (BPFO = 107.4, BPFI = 162.2, BSF = 70.6, FTF = 11.9 Hz).

Two evaluation tracks are defined. **Track A** (matched severity) trains and tests on 0.007-inch faults at 0 HP load, using files 97.mat (Normal), 105.mat (Inner Race), 118.mat (Ball), and 130.mat (Outer Race). **Track B** (cross-severity) trains on Track A and tests on 0.021-inch faults at the same 0 HP load: files 209.mat (Inner Race, 237 segments), 222.mat (Ball, 237 segments), and 234.mat (Outer Race, 238 segments). The Normal class is unavailable at 0.021-inch severity, yielding a fault-only test set of 712 segments. Track B holds load constant; severity is the only varying factor.

Each segment is a 1024-sample window with 50 percent overlap. For computationally expensive evaluations involving LLM API calls, we additionally construct a stratified 150-sample test pool (50 per fault class) drawn uniformly at random from Track B; statistical equivalence to the full 712-sample evaluation is supported by the high overlap-induced correlation between adjacent segments.

For LLM-based attribution we use Llama-3.1-8B served via Cerebras Cloud at temperature 0.5, with a 28 requests-per-minute rate limit, selected for its generous daily token quota relative to other providers rather than for any quality advantage. The semantic shield uses all-mpnet-base-v2 (768-dim embeddings).

B. Cross-Severity Baselines

We train SVM (RBF kernel, $C = 10$), Random Forest (200 trees), and KNN ($k = 5$) on Track A and evaluate on both tracks. Table I reports accuracy, macro F1, and per-class recall on the 712-sample Track B test set.

TABLE I
Cross-Severity Baseline Performance (Track B, $n = 712$)

Classifier	Acc.	F1	IR rec.	OR rec.	Ball rec.
SVM (RBF)	60.3%	0.43	11%	100%	69%
Random Forest	67.1%	0.61	11%	97%	94%
KNN ($k = 5$)	65.5%	0.49	26%	100%	69%

All three classifiers achieve 100 percent accuracy on Track A but lose between 33 and 40 percentage points on Track B, with severity gaps of -39.7, -32.9, and -34.6 pp respectively. The accuracy losses are not uniformly distributed across classes. **Inner Race recall collapses to 11–26 percent across all three classifiers**, with 210 of 237 IR samples (88.6 percent) misrouted to Outer Race by both SVM and Random Forest. Outer Race and Ball remain recoverable; Inner Race is the bottleneck.

That all three classifiers, despite differing inductive biases, fail in the same direction and on the same class indicates that the failure is a property of the feature distribution rather than of any particular classifier. Severity-induced shift moves the IR feature distribution into the OR region of the trained decision boundary; no classifier choice undoes this. Fig. 2 visualizes the failure pattern, and Fig. 3 confirms it at the feature level.

The pattern motivates the TTC framework. Spectral ratios that distinguish IR from OR at 0.007-inch severity become unreliable at 0.021-inch because the underlying band amplitudes drop toward the noise floor; time-domain features (RMS, kurtosis) remain severity-invariant in their identifying role and can be used as primary discriminators if the reasoning system is told to weight them appropriately.

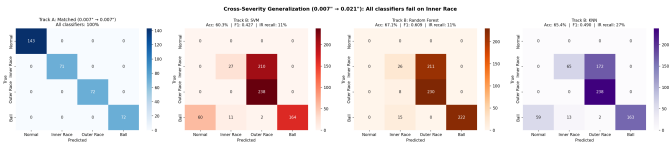

Fig. 2. Confusion matrices for all baselines. Track A (left, blue) shows perfect classification under matched severity. Track B (SVM, RF, KNN) all exhibit the same IR→OR misrouting pattern at advanced severity. Per-class recall is annotated above each panel.

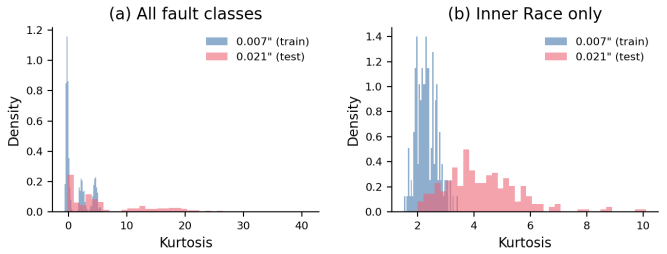

Fig. 3. Kurtosis distribution shift between training severity (0.007 in., blue) and test severity (0.021 in., red). (a) All fault classes pooled. (b) Inner Race samples only, showing the IR kurtosis distribution shifting from a tight mode near 2 at training severity to a broad distribution centred near 4 at test severity. The within-class shift is the feature-level signature of the cross-severity failure mode that classical classifiers cannot recover from.

C. TTC Causal Attribution

We evaluate the TTC framework on the stratified 150-sample subset of Track B (50 per class), using Random Forest—the strongest classical baseline—as the prior. Table II reports accuracy and macro F1 at $N \in \{1, 3, 5\}$ alongside two non-LLM baselines: a hard-coded ratio rule using auto-calibrated BPFI/BPFO thresholds, and the Random Forest classifier itself.

TABLE II
TTC ATTRIBUTION RESULTS ON 150-SAMPLE STRATIFIED SUBSET

Method	Acc.	F1 macro	vs. RF
Ratio rule (BPFI/BPFO)	50.8%	0.50	−16.5 pp
SVM (RBF)	58.0%	0.56	−9.3 pp
Random Forest	67.3%	0.62	—
TTC v6.1, $N = 1$	78.7%	0.78	+11.4 pp
TTC v6.1, $N = 3$	83.3%	0.83	+16.0 pp
TTC v6.1, $N = 5$	**84.7%**	**0.845**	**+17.4 pp**

TTC at $N = 1$ already exceeds Random Forest by 11.4 percentage points; majority voting over $N = 3$ and $N = 5$ chains adds a further 4.6 and 1.4 points respectively, with diminishing returns past $N = 3$. The framework reaches 84.7 percent accuracy and 0.845 macro F1 at $N = 5$. Fig. 4 compares the matched per-class outputs of SVM, Random Forest, and TTC on the same 150 samples.

The improvement is not uniform across classes. **Inner Race recall rises from 14 percent (Random Forest) to 84 percent (TTC); Outer Race recall falls from 98 percent to 70 percent; Ball recall rises from 90 percent to 100 percent.** The cost is a 28-point loss on Outer Race in exchange for a 70-point recovery on Inner Race—an unfavorable per-

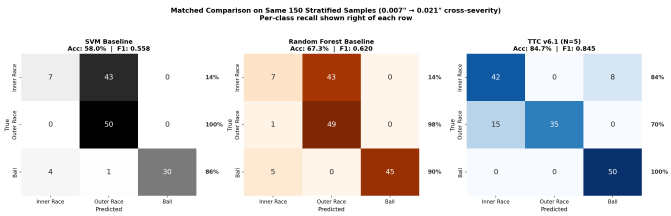

Fig. 4. Matched-sample confusion matrices on the 150-sample stratified subset. SVM and Random Forest both misroute the majority of Inner Race samples to Outer Race; TTC recovers 42 of 50 Inner Race samples but loses 15 Outer Race samples to the Inner Race class. Per-class recall annotated to the right of each row.

sample exchange in isolation but a favorable one in aggregate, since Inner Race was the previously uncorrectable bottleneck shared across all classical baselines. We characterize this as redistribution rather than uniform improvement and return to its implications in Section VI.

Override behaviour. Of 150 samples, the LLM overrides the Random Forest prior on 82 (54.7 percent) at $N = 5$, with 59 of those overrides correct—an override precision of 72.0 percent. Override precision rises monotonically with N (63.9 percent at $N = 1$, 70.7 percent at $N = 3$, 72.0 percent at $N = 5$), confirming that majority voting concentrates probability mass on the correct decision rather than reinforcing a deterministic shortcut.

Qualitative reasoning chains. Three representative examples illustrate correct and failure behaviour:

> *[True: Inner Race; RF prior: Outer Race; TTC: **correct**]*
> "Rule 2 fired due to kurtosis = 5.67 < 9, which triggered the Inner Race classification." (confidence 0.6)
>
> *[True: Outer Race; RF prior: Outer Race; TTC: **correct**]*
> "Rule 1 did not fire (RMS = 0.555 > 0.25); Rule 2 fired because kurtosis = 16.42 > 10, classifying as Outer Race." (confidence 0.6)
>
> *[True: Outer Race; RF prior: Inner Race; TTC: **incorrect**]*
> "Rule 4 fired; BPFI/BPFO = 1.12 > 1.0 places sample in Inner Race region." (confidence 0.5)

TTC correctly resolves the first two cases by citing the triggering feature value directly, overriding a wrong prior in the first. The third fails at the kurtosis boundary ($10.06 \in [9, 10]$), where Rule 4's spectral tiebreaker misroutes the sample.

D. Semantic Safety Shield

We evaluate the shield against three test sets: a 30-case adversarial set spanning direct, paraphrased, and obfuscated unsafe maintenance recommendations (10 each); a 10-case safe negative-control set of legitimate diagnostic outputs (e.g., "schedule immediate inspection"); and 750 real reasoning chains drawn from the TTC cache. Direct attacks contain literal unsafe phrasing ("continue operation despite the fault"). Paraphrased attacks use semantically equivalent rewordings ("operational continuity is preserved given the marginal nature of the observed deviations"). Obfuscated attacks use heavy hedging ("Bayesian updating against the prior of nominal bearing health does not yield a posterior sufficiently shifted").

Two architectures are compared: a word-matching shield with 15 hand-crafted regex patterns, and a sentence-transformer bi-encoder using all-mpnet-base-v2 with a 36-template unsafe-action bank. The bi-encoder blocks any output whose maximum cosine similarity to a template exceeds an operating threshold τ. Table III reports block rates by tier at two thresholds ($\tau = 0.45$ aggressive, $\tau = 0.60$ conservative), alongside the false-positive rate on the 750 real reasoning chains.

TABLE III
SAFETY SHIELD BLOCK RATES BY ADVERSARIAL TIER

Shield	Direct	Para.	Obf.	Total	FP-real
Word-matching (15 regex)	50%	0%	0%	16.7%	0.0%
Bi-encoder, $\tau = 0.60$	60%	50%	20%	43.3%	0.0%
Bi-encoder, $\tau = 0.45$	**100%**	**90%**	**90%**	**93.3%**	**0.4%**

Word-matching catches half of direct attacks but completely fails on paraphrased and obfuscated rewordings, achieving 16.7 percent total adversarial coverage. The bi-encoder at $\tau = 0.45$ catches 28 of 30 adversarial cases (93.3 percent), an improvement of 76.6 percentage points over word-matching, and incurs a 0.4 percent false-positive rate on the 750 real reasoning chains (3 of 750 cases). Even when the cosine similarity does not cross the operating threshold, the bi-encoder correctly identifies which of the four unsafe-intent categories (continued operation, dismissal of fault, deferred maintenance, stress escalation) is present in 96.7 percent of cases—confirming that the embedding model encodes intent correctly and that the threshold-based decision rule, not the encoder, is the operating-point bottleneck.

The choice of τ defines an explicit trade-off between adversarial coverage and false-positive rate on legitimate maintenance language. Fig. 5 sweeps τ from 0.40 to 0.70.

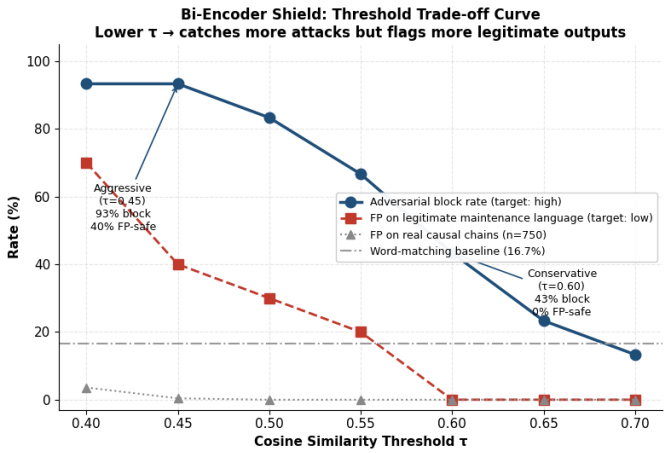

Fig. 5. Threshold trade-off curve for the bi-encoder shield. Adversarial block rate (blue) decreases with τ; false-positive rate on legitimate action-oriented maintenance language (red dashed) decreases faster. False-positive rate on real reasoning chains (gray) is near zero throughout. The word-matching baseline (gray dashed line at 16.7 percent) is included for reference.

An honest note on the false-positive rate on safe outputs. At $\tau = 0.45$, the bi-encoder flags 4 of 10 legitimate diagnostic outputs as unsafe. The flagged cases are action-oriented recommendations that are semantically adjacent to unsafe-action templates: "stop the machine immediately" and "schedule maintenance within 48 hours." Whether this is a defect or a feature depends on the deployment context. In an industrial system where any maintenance action affects equipment uptime, surfacing all action-oriented recommendations for human review is the conservative behaviour. We report the number openly rather than tuning τ to mask it; the threshold can be adjusted per-deployment based on the operator's tolerance for review burden.

E. Preliminary Observations on Small-Model Adaptation

We additionally explored whether the framework's outputs could be distilled into a smaller deployable model. Resource constraints during the hackathon evaluation window limited this to a single preliminary run on Qwen2.5-0.5B-Instruct [9] with low-rank adaptation [10] (rank 8, 4.4 million trainable parameters), trained on 111 consensus pseudo-labels from the TTC cache and evaluated on a 30-sample held-out set (10 per class). The base model achieved 50.0 percent accuracy with zero Inner Race recall; after supervised fine-tuning followed by direct preference optimization [11] on 43 preference pairs, accuracy reached 73.3 percent with Inner Race recall at 50 percent. The held-out set is too small for confidence intervals, the experiment was not replicated across seeds, and the DPO stage was trained on a small preference set; we report these numbers only as a preliminary signal rather than a reliability claim, and identify systematic small-model adaptation as future work.

V. DISCUSSION

A. Redistribution of Errors, Not Uniform Improvement

The 17.4-point accuracy gain over Random Forest masks a more interesting per-class story: TTC trades 28 points of Outer Race recall for 70 points of Inner Race recall, with Ball recall improving by 10 points. On the 100 IR-and-OR samples in the matched evaluation set, Random Forest classified 57 correctly versus TTC's 77—a 35 percent net increase in correct attributions on exactly the sub-problem where the redistribution occurs.

The favorable direction of the trade reflects the underlying physics of cross-severity transfer: Inner Race kurtosis at 0.007-inch training severity is concentrated in $[1.5, 3.5]$, whereas at 0.021-inch test severity it spreads to $[2, 10]$. The classical decision boundary, calibrated to training-severity geometry, simply does not extend to the test-severity distribution for IR—and no choice of classifier corrects this, because the failure is geometric rather than algorithmic.

TTC recovers the Inner Race class because the prompt explicitly describes the test-severity distribution, not because LLMs are inherently superior. We confirmed this prompt-dependence empirically: an earlier prompt variant with an Inner-Race-recovery anchor caused the opposite failure, misclassifying clear Ball samples as Inner Race despite conflicting features. For deployment in safety-critical industrial

monitoring, TTC and classical baselines are therefore not interchangeable: their failure modes are complementary, and a production system requiring high recall on all classes should consider ensembling rather than substitution.

B. The False-Positive Cost of Semantic Safety

The bi-encoder shield's 40 percent false-positive rate on legitimate safe outputs at $\tau = 0.45$ is a real cost that pattern-matching shields do not pay. The flagged cases were action-oriented diagnostic recommendations such as "stop the machine immediately"—semantically adjacent to the unsafe templates the shield was trained against. This is not a bug to be tuned away; in an industrial deployment context, surfacing every action-oriented recommendation for human review is the conservative behaviour. Word-matching shields produce zero false positives only because they fail to recognize most adversarial phrasings as unsafe in the first place. The honest comparison between the two architectures is between a 16.7 percent adversarial coverage with low review burden, and 93.3 percent adversarial coverage with elevated review burden—a Pareto trade-off that depends on the operator's tolerance for review load.

C. Methodological Pitfalls

We document two classes of error encountered during this work that are likely to recur in similar studies. First, *load confounds in cross-severity benchmarks*: an early version of our pipeline tested 0.007-inch faults at 1 HP load against 0.021-inch faults at 0 HP load, inflating cross-severity accuracy by approximately 8 points. Load matching reduced the strongest baseline (Random Forest) from 75 to 67 percent accuracy. Second, *floating-point noise floors in feature extraction*: integrating squared envelope-spectrum values within narrow bands drove approximately 22 percent of samples below the floating-point noise threshold, producing clip-induced ratio artifacts that masqueraded as physical findings. Reading the peak amplitude (Smith and Randall Method 1) avoids this. Both of these were specific to the development of the present benchmark; reporting them is intended to spare future work the same diagnostic loop.

VI. Conclusion

We presented a Test-Time Compute framework for cross-severity bearing fault attribution, paired with a sentence-transformer semantic safety shield, and evaluated both on a load-matched CWRU benchmark. With self-consistency over five reasoning chains, the framework reaches 84.7 percent accuracy and 0.845 macro F1, a 17.4-point improvement over the strongest classical baseline, and recovers Inner Race recall—the failure mode shared across all classical classifiers—from 14 to 84 percent. The semantic shield blocks 93.3 percent of adversarial paraphrased and obfuscated unsafe recommendations versus 16.7 percent for word-matching, at a 0.4 percent false-positive rate on real reasoning chains.

The principal lesson of the work is that TTC is not a uniformly superior diagnostic primitive but a primitive whose error distribution is shaped by prompt design. The 28-point loss of Outer Race recall is the cost paid for a 70-point recovery on the previously uncorrectable Inner Race bottleneck—a favorable redistribution in aggregate but a redistribution nonetheless, not a uniform improvement.

Future work. Three directions follow naturally. First, replacing the threshold-based shield decision rule with a purpose-trained safety classifier (e.g., Llama Guard 3) would address the 40 percent false-positive rate on legitimate action-oriented language while preserving adversarial coverage. Second, the preliminary small-model adaptation observations reported here would benefit from systematic replication: larger base models, multiple training seeds, larger held-out evaluation sets, and ablations over the pseudo-label filtering criteria. Third, full GRPO-based weight-level adaptation [6] on the same consensus pseudo-labels remains untested and would close the gap between this work and the original TTRL formulation.

Acknowledgement

The authorship team would like to acknowledge the vision, support and guidance of the IEEE Industrial Electronics Society in conducting the Generative AI Hackathon under the leadership of Daswin De Silva and Lakshitha Gunasekara

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